

Effects of Conditional Cash Transfers on School Dropout Rates: A Bayesian Approach to Modelling Linear Regression

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1 Abstract

This report presents an analysis of data modeled using a Bayesian approach to linear regression. The main goal was to understand the effects of the educational social program, *Prospera*, on dropout rate in Mexican's most marginalized schools. The main result driven from the analysis is that cash transfers provided by the program poorly improves dropout rates. The first section presents background information of the social program and data. The analysis section presents the Bayesian approach, like details on the computation of the posterior distributions and checks over model and priors implemented. Finally interpretations over results are provided.

2 Social program Prospera and data

The Mexican government has adopted social programs that seek to address the main social concerns in the country. Specifically for education, the program Prospera was adopted in 1988, with the name of *Solidaridad*, although the program's name has varied through the years. Nowadays, it is runned with the name of *Bienestar*, but we will stick to the name ***Prospera*** in this report, since data analyzed correspond to this stage of the program. This social program acts through Conditional Cash Transfers, where mothers are provided with the aid conditioned on having their children attend to school.

One of the educational main concerns in Mexico is that students drop out from schools. This is why the main goal of this analysis is to understand how Prospera has impacted over dropout rates

in Mexican schools. Specially in schools characterized by the Mexican Population Council as the most marginalized schools from the country.

Data was provided by Dr. Francisco Cabrera-Hernández, from the Center for Research and Economics Teaching. Units in the dataset are at a school level and the presented analysis uses 79,984 schools from 2006 to 2018.

3 Analysis

The main goal of this project is to address estimations through Bayesian modeling. The approach taken is the regression line. Thus, for the goal of the analysis explained in the past section, the regression line considered is:

$$Y_i = \beta_0 + \beta_1 \text{treatment}_i + \beta_2 \text{nteacher}_i + \beta_3 \text{nteacher}_i^2 + \beta_4 \text{teacherbcs}_i + \beta_5 \text{csize}_i + \epsilon_i. \quad (1)$$

Where for school i , Y_i is the percentage of total drop out and β_1 is the parameter of interest. The treatment variable takes value 1 when at least 50% of the students in school i are provided with cash transfer and 0 otherwise. Other control variables are added, such as number of teachers as a polynomial of second order, proportion of teachers with bachelor's degree (as a proxy for average education of teachers in schools) and size of classes.

Considering the Bayesian approach on the regression model, the goal is to address the posterior distributions for the parameters β and σ^2 , that correspond to the vector of coefficients and the variance of the error term, respectively.

The first step is to assign prior distributions for the parameters. The decision taken was to use non informative priors. The primary reason is that the analysis focuses on a social program, where objectivity is essential. Also, even though *Prospera* is a very studied program, results in studies tend to vary significantly due to the complexity of model specification in educational studies. Moreover, political interests and a multitude of different analytical approaches exert influence on the diverse studies conducted on the social program. Therefore the decision for this analysis was not to implement previous information.

With this in mind, unit information priors were imposed to the parameters. In particular

$$\beta \sim N_p(\beta_0, \Sigma_0)$$

and

$$\sigma^2 \sim IG\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right).$$

The goal is to “let information speak for itself”, therefore this is achieved through these priors which considers information given only by one single observation thanks to the parameters chosen for the prior distributions. These are $\beta_0 = \hat{\beta}_{ols}$ and $\Sigma_0 = \frac{\mathbf{X}'\mathbf{X}}{n\sigma_{ols}^2}$. Note that OLS is the main source of information for priors.

Prior predictive checks were performed to analyse the unformativeness of the priors implemented. To address this, 200 distributions for the outcome were replicated with parameters drawn from the prior distributions.

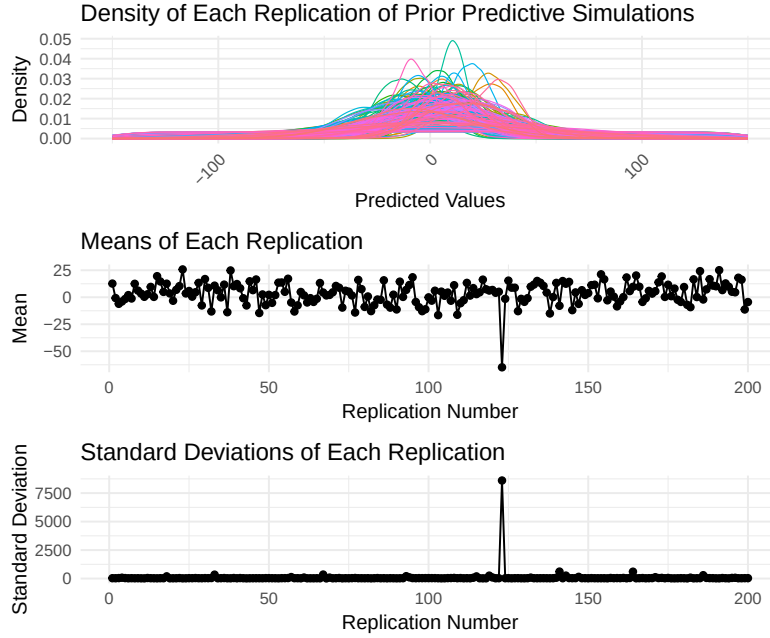


Figure 1: 200 replicated data, means and standard deviations.

Plots from Figure 1 express the unformativeness of the prior distributions. The first plot exposes all of the 200 y^{rep} distributions. Unformativeness is clear due to the gaussian form of the distributions and means around 0, suggesting no relevant skews. The second and third plot shows important statistics: the mean and standard deviations of the 200 replicated data. These statistics are centered around 0, in the case of means, and remain short and constant in the case of standard deviations. An important remark are the single extreme observations in the statistics. We

can assume these are due to possible outliers but this do not change the overall tendencies in the information provided by priors.

Therefore, we can conclude a clear lack of information that the prior distributions provide. This means our goal to estimate posterior distributions for $\theta = (\beta, \sigma^2)$ will be heavily influenced by the model stated and less likely by the priors. Objectiveness, then, is achieved.

Continuing with the analysis, is essential to set a model for the outcome variable to then compute posterior distributions. Take into account that Y_i , without considering uncertainty, has a linear form conditional on the covariates. This is:

$$E[Y|\mathbf{x}] = \beta_1 + \dots + \beta_p x_p = \beta^T \mathbf{x}$$

but since $\epsilon_i \sim N(0, \sigma^2)$ then a model for Y_i that provide a complete specification conditional on covariates, vector of coficients and variance in error is the multivariate normal distribution:

$$\begin{aligned} p(y_1, \dots, y_n \mid x_1, \dots, x_n, \beta, \sigma^2) &= \prod_{i=1}^n p(y_i \mid x_i, \beta, \sigma^2) \\ &= (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta^T x_i)^2 \right\}. \end{aligned}$$

where $\sum_{i=1}^n (y_i - \beta^T x_i)^2$ is the $SSR(\beta)$.

Now that priors and model are set, the goal is to compute posterior distributions for θ . The approach taken was a Markov Chain Monte Carlo Method (MCMC): Gibbs Sampling. This algorithm approximates the joint posterior distribution $p(\beta, \sigma^2 | y, X)$. The full conditional distributions used are:

$$p(\beta | y, X, \sigma^2) = N_p \left(\left(\Sigma_0^{-1} + \frac{\mathbf{X}'\mathbf{X}}{\sigma^2} \right)^{-1} \left(\Sigma_0^{-1} \beta_0 + \frac{\mathbf{X}'\mathbf{y}}{\sigma^2} \right), \left(\Sigma_0^{-1} + \frac{\mathbf{X}'\mathbf{X}}{\sigma^2} \right)^{-1} \right)$$

and

$$p(\sigma^2 | y, x, \beta) = InvGamma \left(\frac{\nu_0 + n}{2}, \frac{\nu_0 \sigma_0^2 + SSR_\beta}{2} \right)$$

.

In particular the Gibbs sampler works as follows: given current values $\{\beta^{(s)}, \sigma^{2(s)}\}$ that in our case are $\{\hat{\beta}_{ols}^{(s)}, \hat{\sigma}_{ols}^{2(s)}\}$, new values are generated through

1. updating β :

(a) compute $\mathbf{V} = \text{Var}[\beta \mid \mathbf{y}, \mathbf{X}, \sigma^{2(s)}]$ and $\mathbf{m} = \mathbb{E}[\beta \mid \mathbf{y}, \mathbf{X}, \sigma^{2(s)}]$

(b) sample $\beta^{(s+1)} \sim \text{multivariate normal}(\mathbf{m}, \mathbf{V})$

2. updating σ^2 :

(a) compute $\text{SSR}(\beta^{(s+1)})$

(b) sample $\sigma^{2(s+1)} \sim \text{inverse-gamma}\left(\frac{\nu_0+n}{2}, \frac{\nu_0\sigma_0^2+\text{SSR}(\beta^{(s+1)})}{2}\right)$

with an algorithm large enough to approximate the joint posterior distribution.

After running the algorithm for 5000 scans, Table 1 shows the OLS estimations and means from the posterior distributions for all parameters on the regression model.

Table 1: Comparison of OLS Regression Estimates and Means of Posterior Distributions from Gibbs Sampling

Parameter	OLS Estimate	Mean of Posterior Distribution
(Intercept)	3.6567	3.6576
cash transfer	-1.1662	-1.1674
nteacher	-0.5894	-0.5890
nteacher ²	0.0275	0.0275
teachratio	0.5090	0.5089
teacherbcs	-0.0186	-0.0186
csize	0.1274	0.1274

Table 1 suggests, first, that due to the uninformativeness of the priors and the parameters imposed, the Gibbs sampling outcome is very similar to the OLS estimation. In fact, Figure 2 has plots of all posterior densities per parameter, where red line is the mean of the posterior distributions and the blue line the OLS estimations. In fact, these lines are perfectly overlapped.

The second important thing to highlight is that the mean of posterior distributions correspond to $\hat{\beta}_{Bma} = \sum_s \beta^{(s)} / S$. Thus, we can obtain predicted values since $\hat{y}_{test} = \mathbf{X} \hat{\beta}_{Bma}$. Although the main purpose of the analysis is to evaluate the actual impact of treatment on the outcome variable and not predictions.

The interpretation of posterior densities on Prospera will be described in the following section. Nevertheless, before doing so, it's important to first analyse the model implemented for the outcome to understand every element used in the Bayesian update. Thus, model checking was performed by sampling from the posterior distributions to generate predicted values for the outcome, denoted as

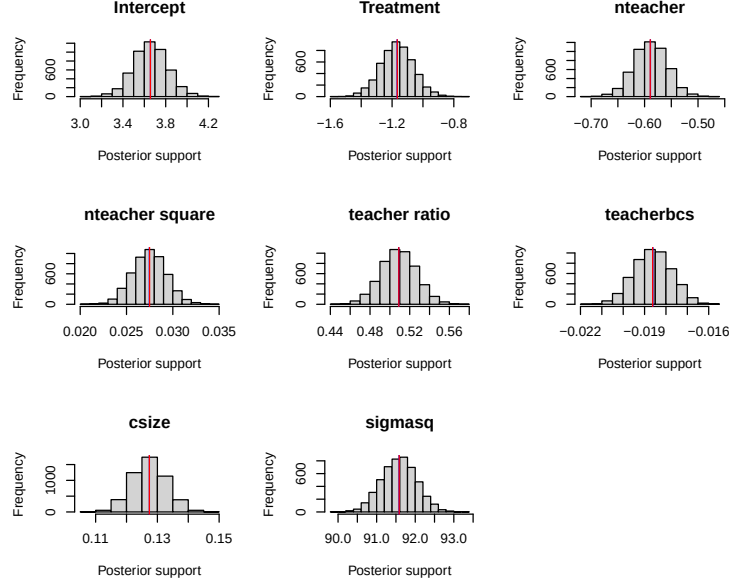


Figure 2: 200 replicated data, means and standard deviations.

y^{rep} . The purpose of doing so is to analyse if the model is able to replicate crucial summaries from observed data, denoted as y^{obs} .

When performing model checking, test statistics (or discrepancy measures) are very helpful for comparing data. Therefore, this tool was used to compare a summary of replicated data and observed data. In this case, the goal is to answer if modeling Y_i through a multivariate normal distribution is able to replicate, for example, the mean of the observed data. Thus, consider the test statistic $T(y)_{obs} = \sum_{i=1}^n y_i/n$ against the same but for y^{rep} . The first plot of Figure 3 shows this. Note that the distribution of means from y^{rep} are centered in the mean of y^{obs} , blue and red line respectively. In fact, the mean of test quantities $T(y^{rep})$ is 4.004299 and $T(y^{obs}) = 4.003026$. This suggest a normal multivariate distribution is clearly an appropriate model to capture the mean of y^{obs} . Remind that data has almost 80,000 units, so capturing $T(y^{obs})$ is crucial.

For model checking, Posterior Predictive p -values (PPP) can be used to characterize models implemented in Bayesian analysis. In fact, the PPP is defined as the probability that the replicated data could be more extreme than the observed data, this measured with the test quantities is:

$$p_B = Pr(T(y^{rep}, \theta) \geq T(y, \theta) | \theta).$$

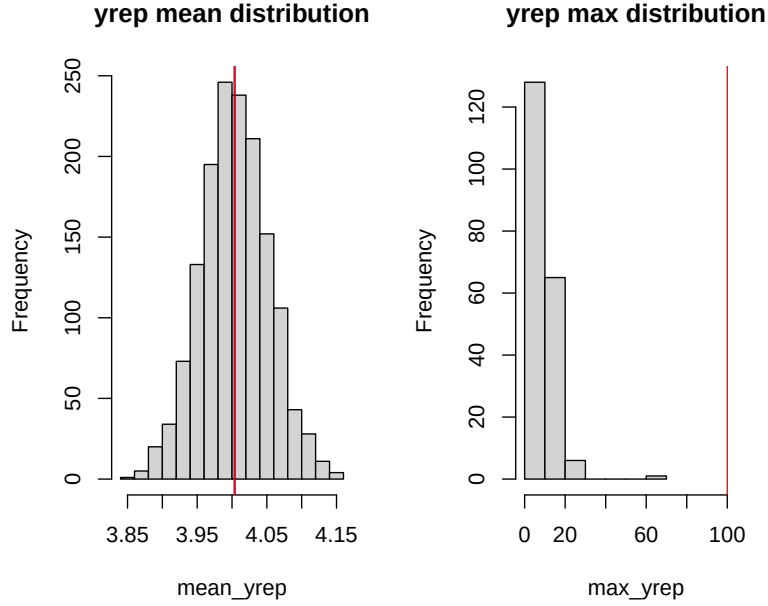


Figure 3: Distributions of $T(y^{rep})$, means and max values respectively.

This implies that a p-value around 0.5 represents an appropriate model implemented. For the case considered before where $T(y)_{obs} = \sum_{i=1}^n y_i/n$, the $p_B = 0.5006667$, which demonstrates that the model implemented captures y^{obs} mean successfully. However, this summary of data is not the only one that is of interest. Figure 4 displays the actual distribution of the outcome variable. The plot shows that y^{obs} contains extreme values outside the centered distribution. This generates an important concern to consider if the normal multivariate model is able to capture this characteristic. To address this, $T(y) = \max|y_i|$ was analysed. The second plot of Figure 3 plots the distribution of $T(y^{rep}) = \max|y_i|$. The red line that represents $T(y^{obs}) = \max|y_i| = 100$ is also expressed. For this case $p_B = 0$, thus the model is not able to capture extreme values from the observed data at all. Although this is not critical since outliers values in y^{obs} are just a few.

After all this discussion, now all elements behind the computation of the posterior distribution have been checked. This includes the priors used, the algorithm executed for the posterior distributions and the model implemented. Therefore the following section continues with the interpretation of results over the object of analysis in this report.

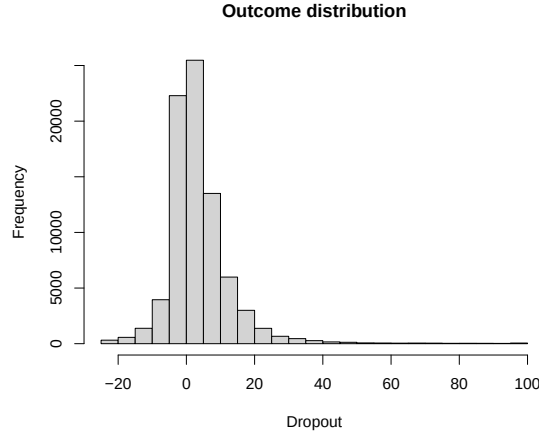


Figure 4: Distribution of dropout in schools.

4 Interpretation and conclusions

Consider again the goal of the analysis: how do Prospera’s cash transfers affect drop out rates in Mexican’s most marginalized schools? Before stating on this, it’s important to remark that results presented in further lines are not causal effects since the scoop of this report is limited only to the theory behind the Bayesian approach on linear regression and, unfortunately, fundamental analysis of endogeneity, heteroskedasticity, common support, variability over time and other robustness checks are not implemented. Although for the sake of interpreting estimations, this report presents the effects of covariates over outcome.

Table 2: Posterior Means and 95% Credible Intervals for Betas and σ^2

	Posterior Mean	2.5%	97.5%
(Intercept)	3.6576	3.3366	3.9713
Cash transfer	-1.1674	-1.3723	-0.9507
Num. teachers	-0.5890	-0.6498	-0.5270
Num. teachers ²	0.0275	0.0238	0.0311
Teacher ratio	0.5089	0.4740	0.5439
Teacher with bc’s	-0.0186	-0.0203	-0.0170
Class size	0.1274	0.1163	0.1384
σ^2	91.59279	90.70267	92.46518

That being said, results are shown in Table 2. The parameter of interest is β_1 from (1). It's possible then to conclude that cash transfers have a small but negative effect on dropout rates in Mexican's most marginalized schools. This indicates that Prospera's performance to improve education outcomes, such as reducing dropout rates, has been poor at least from 2006 to 2018. Effects of the others covariates vary but remain small, even more than the treatment effect. With respect to the Bayesian implementation, we can conclude that prior predictive check suggest a correct implementation of priors to archive uniformativeness so data can speak for itself. Also posteriors distributions computed align with theory and most importantly, remain objective. Finally posterior predictive checks insinuate that the model used is able to capture the mean from observed data, which is a fundamental statistic since the data used is large enough to highlight this concern. Also, the few non-captured outliers are not considered to be relevant for the overall result of the posterior computation.